Data Base Management System(DBMS)

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Practical Assignment 8

Practical: Transaction Control Commands (COMMIT, ROLLBACK, SAVEPOINT)

Question: A college maintains student fee payment details in a table named `StudentFees(StudentID, StudentName, TotalFees, PaidFees, PendingFees)`.

Perform the following operations using MySQL TCL commands:

1. Create the `StudentFees` table and insert records for five students.

```
mysql> create table StudentFees(StudentID int Primary Key,StudentName varchar(30),TotalFees int,PaidFees int,PendingFees int);
Query OK, 0 rows affected (0.03 sec)

mysql> desc StudentFees;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| StudentID   | int         | NO   | PRI | NULL    |       |
| StudentName | varchar(30) | YES  |     | NULL    |       |
| TotalFees   | int         | YES  |     | NULL    |       |
| PaidFees    | int         | YES  |     | NULL    |       |
| PendingFees | int         | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.03 sec)
```

```
mysql> select * from StudentFees;
+-----------+---------------+-----------+----------+-------------+
| StudentID | StudentName   | TotalFees | PaidFees | PendingFees |
+-----------+---------------+-----------+----------+-------------+
|      1001 | Anurag Lad    |    100000 |    60000 |       40000 |
|      1002 | Harshad Teli  |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki |    100000 |    90000 |       10000 |
|      1004 | Neha Sutar    |    100000 |    70000 |       30000 |
|      1005 | Anjali Patil  |    100000 |    65000 |       35000 |
+-----------+---------------+-----------+----------+-------------+
5 rows in set (0.00 sec)

mysql>
```

2. Start a transaction and update the fee payment of the first student by ₹10,000.

```
mysql> update StudentFees set PaidFees=PaidFees+10000,PendingFees=PendingFees-10000 where StudentID=1001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentFees;
+-----------+---------------+-----------+----------+-------------+
| StudentID | StudentName   | TotalFees | PaidFees | PendingFees |
+-----------+---------------+-----------+----------+-------------+
|      1001 | Anurag Lad    |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli  |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki |    100000 |    90000 |       10000 |
|      1004 | Neha Sutar    |    100000 |    70000 |       30000 |
|      1005 | Anjali Patil  |    100000 |    65000 |       35000 |
+-----------+---------------+-----------+----------+-------------+
5 rows in set (0.00 sec)

mysql>
```

3. Create a savepoint named `SP1`.

```
mysql> SAVEPOINT S1;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

4. Update the fee payment of the second student by ₹15,000.

```
mysql> update StudentFees set PaidFees=PaidFees+15000,PendingFees=PendingFees-15000 where StudentID=1002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    65000 |       35000 |
|      1003 | Vardhan Patki|    100000 |    90000 |       10000 |
|      1004 | Neha Sutar   |    100000 |    70000 |       30000 |
|      1005 | Anjali Patil |    100000 |    65000 |       35000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)

mysql>
```

5. Create another savepoint named `SP2`.

```
mysql> SAVEPOINT SP2;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

6. Update the fee payment of the third student by ₹20,000.

```
mysql> update StudentFees set PaidFees=PaidFees+20000,PendingFees=TotalFees-PaidFees where StudentID=1003;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki|    100000 |   100000 |           0 |
|      1004 | Neha Sutar   |    100000 |    75000 |       25000 |
|      1005 | Anjali Patil |    100000 |    65000 |       35000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)
```

7. Roll back to savepoint `SP1` to cancel the changes made after it.

```
mysql> ROLLBACK TO S1;
Query OK, 0 rows affected (0.00 sec)
```

8. Update the fee payment of the fourth student by ₹5,000.

```
mysql> update StudentFees set PaidFees=PaidFees+5000,PendingFees=PendingFees-5000 where StudentID=1004;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki|    100000 |    90000 |       10000 |
|      1004 | Neha Sutar   |    100000 |    75000 |       25000 |
|      1005 | Anjali Patil |    100000 |    65000 |       35000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)
```

9. Display the updated fee payment records.

```
mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki|    100000 |   100000 |           0 |
|      1004 | Neha Sutar   |    100000 |    75000 |       25000 |
|      1005 | Anjali Patil |    100000 |    65000 |       35000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)
```

10. Use `COMMIT` to permanently save the remaining changes.

```
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

11. Start a new transaction and update the fee payment of the fifth student by ₹8,000.

```
mysql> Start Transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> update StudentFees set PaidFees=PaidFees+8000,PendingFees=TotalFees-PaidFees where StudentID=1005;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki|    100000 |   100000 |           0 |
|      1004 | Neha Sutar   |    100000 |    75000 |       25000 |
|      1005 | Anjali Patil |    100000 |    73000 |       27000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)

mysql>
```

12. Use `ROLLBACK` to cancel the last transaction.

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

13. Display all records and verify which changes were saved and which were cancelled.

```
mysql> select * from StudentFees;
+-----------+--------------+-----------+----------+-------------+
| StudentID | StudentName  | TotalFees | PaidFees | PendingFees |
+-----------+--------------+-----------+----------+-------------+
|      1001 | Anurag Lad   |    100000 |    70000 |       30000 |
|      1002 | Harshad Teli |    100000 |    50000 |       50000 |
|      1003 | Vardhan Patki|    100000 |   100000 |           0 |
|      1004 | Neha Sutar   |    100000 |    75000 |       25000 |
|      1005 | Anjali Patil |    100000 |    65000 |       35000 |
+-----------+--------------+-----------+----------+-------------+
5 rows in set (0.00 sec)

mysql>
```